DRAW OR NOISE?

CHRISTY QUANG — PROJECT 08A: Drawing Machines Midterm

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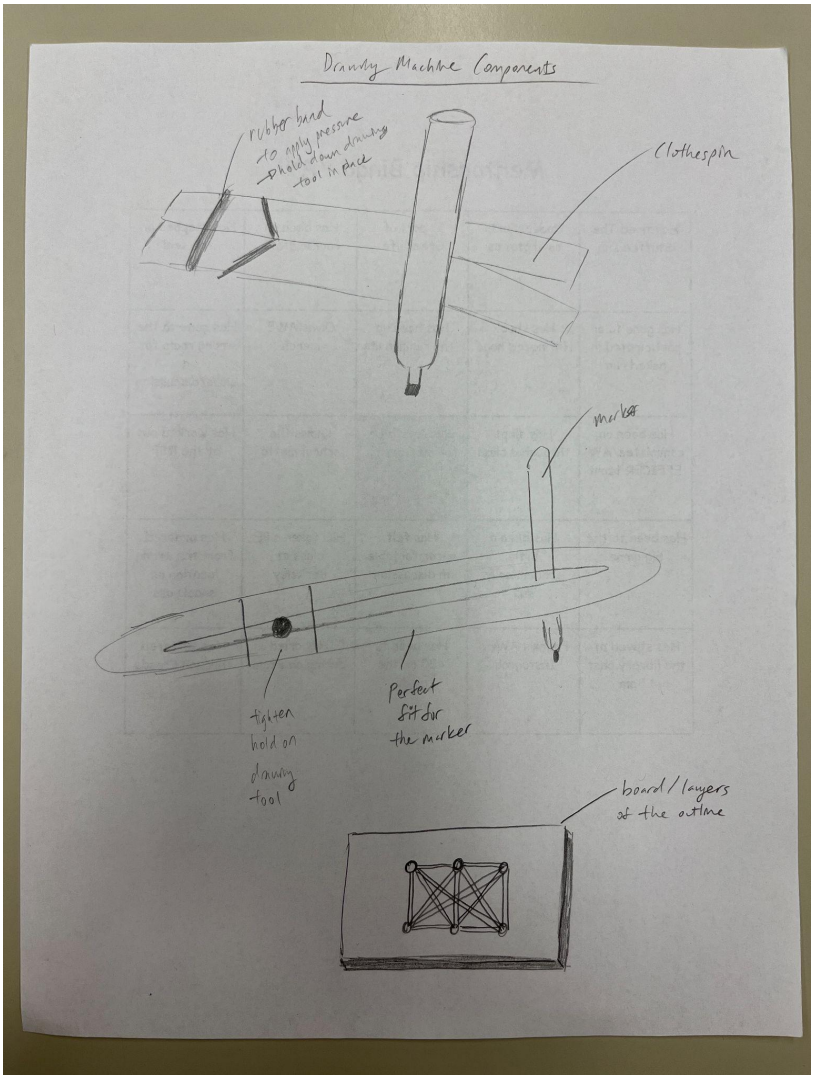
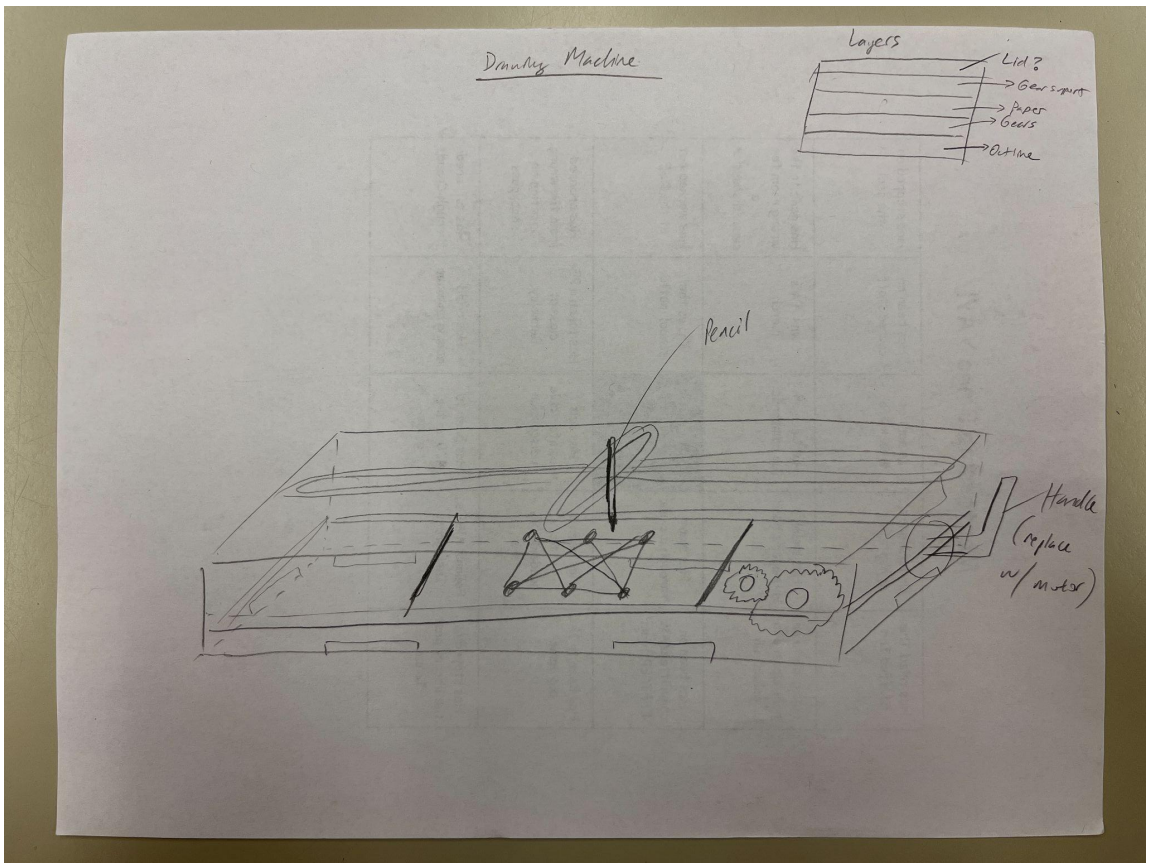
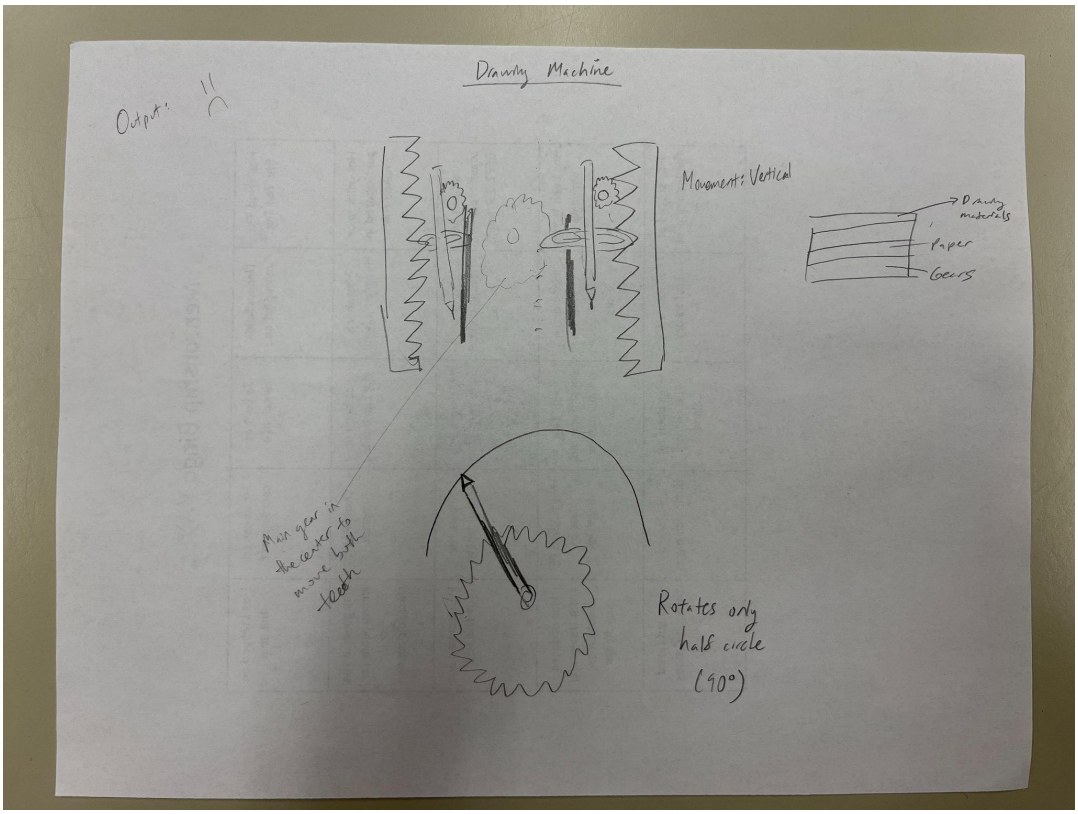
INTRODUCTION

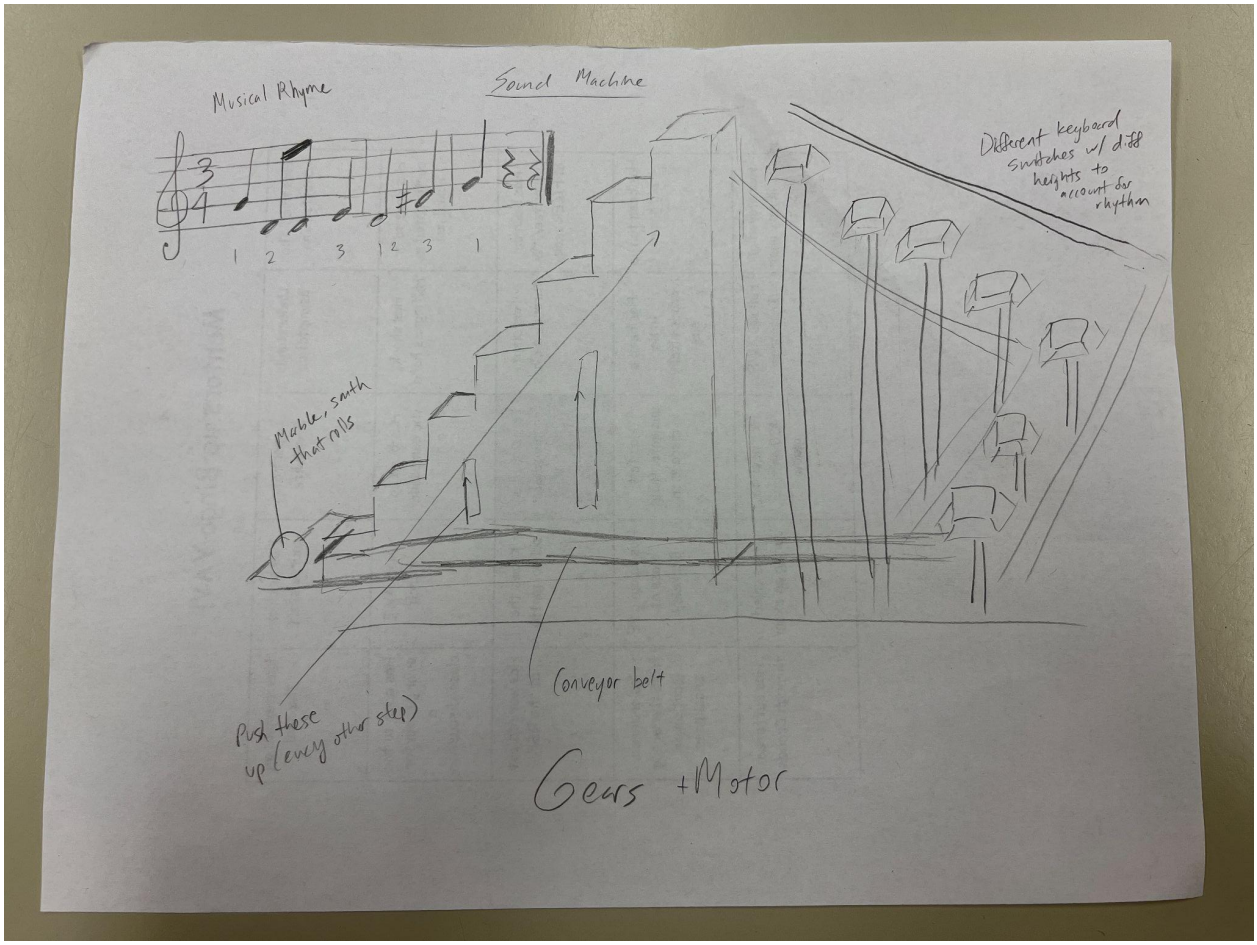
After brainstorming on what to design for either the drawing or sound machine, I decided to create a sound machine because I want to incorporate a sound that has been stuck in my head for a long time. I brainstormed for the drawing machine and while it is intriguing, I want to create the sound machine more because I enjoy making noise.




Drawing Machine
Layers
Lid?
Gear support
Paper
Gears
Outline
Pencil
Handle
(replace
w/ motor)

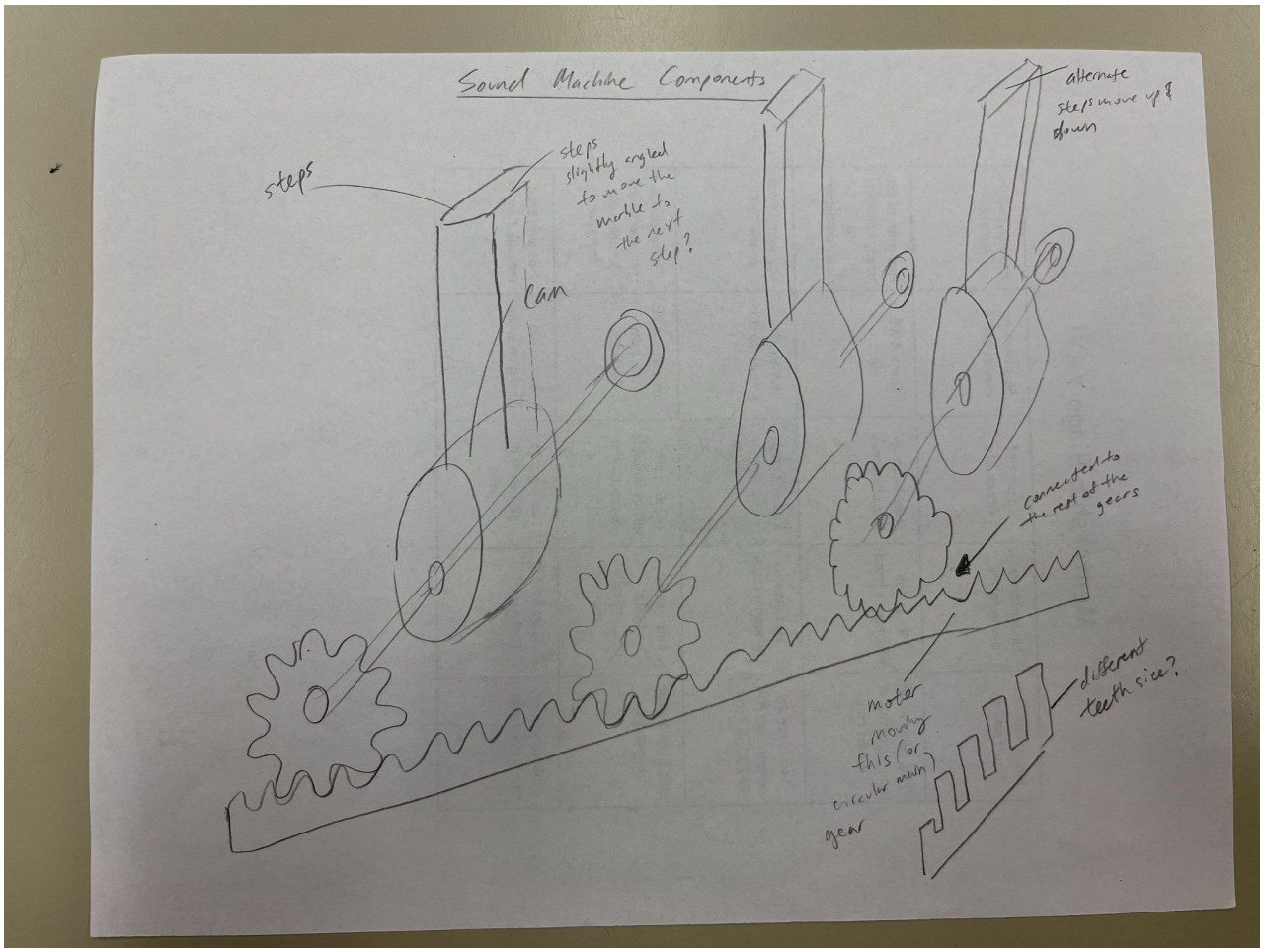

Drawing Machine
Output:
Movement: Vertical
Drawing materials
Paper
Gears
Mini gear in the center to move both teeth
Rotates only half circle (90°)

SOUND MACHINE



Overall Machine

The overall machine involves a staircase that pushes marbles up each step. At the top of the staircase, gravity and the marble's spherical shape causes it to roll down and hit different keyboard switches. The switches are placed at different heights to account for rhythm changes in the musical melody I want to "play". There will be a barrier/railing to help guide the marble down to the conveyor belt where the marble is returned to the staircase and the process is repeated again. The motor to power everything will be hidden inside the staircase.



Mechanism

There will be different sized cams that move together for every other staircase, allowing the marbles to go “up” the steps. To get all of the cams moving at the same time, there will be a motor moving one gear that moves either a tooth rack or all of the other gears together. Once at the top of the stairs, the switches will be placed at different heights and spacing to create the musical pattern. The flexible conveyor belt material will help push the marble back to the staircase where they will go up the staircase once again.

Intended Output

The intended output for this machine is to have the marbles smoothly cycle through the staircase while hitting each switch to create the musical rhythm. Since the rhythm is “long”, I anticipate needing heavier marbles to be able to hit the switches but I’m afraid that will cause the conveyor belt portion to be too weak to support the additional weight. I’m also concerned about the

height of the staircase and how I will align the different switches since I want gravity to help hit each switch and am unsure of how tall I need the staircase to be.